

Invited Perspective: More Greenspace, Less Cognitive Decline? Current Evidence and Future Directions

Li-Xin Hu¹ and Bo-Yi Yang¹

¹Joint International Research Laboratory of Environment and Health, Ministry of Education, Department of Occupational and Environmental Health, School of Public Health, Sun Yat-sen University, Guangzhou, China

<https://doi.org/10.1289/EHP14915>

Refers to <https://doi.org/10.1289/EHP13588>

Alzheimer's disease and related dementias (ADRD) has emerged as a public health concern, with the situation becoming more severe with the aging of the global population.¹ Cognitive deterioration often precedes ADRD,² and the pathological progression of cognitive decline may commence prior to old age.³ Thus, identifying modifiable risk factors of cognitive decline and implementing targeted prevention strategies earlier in life are of great public health significance. Greenspace, as a key component of our living environment, has been beneficially linked to many health outcomes,^{4,5} including improved cognitive function in both childhood⁶ and adulthood.⁷ If this relationship is validated, greenspace could be a useful component of early intervention strategies or therapies to mitigate cognitive decline.

In this issue of *Environmental Health Perspectives*, Jimenez et al.⁸ provide novel evidence concerning the associations between residential greenspace at midlife and cognitive decline later in life among US women with a mean age of 74.2 y at baseline. They also explored greenspace influence stratified by characteristics—namely, socioeconomic status, population density, and apolipoprotein E (APOE-ε4) status—that may make individuals particularly sensitive to cognitive decline. They found that greater exposure to residential greenspace at midlife, measured by the normalized difference vegetation index (NDVI), was associated with better cognitive assessment scores at baseline and a decelerated rate of cognitive decline thereafter, albeit no association was observed between greenspace and verbal memory. Specifically, each interquartile range (IQR) increase in NDVI within a 270-m buffer of residence was estimated to delay cognitive aging by ~8 months.⁸ Moreover, they reported that the beneficial associations were stronger among women residing in neighborhoods with lower socioeconomic status and higher population density, as well as among those carrying the APOE-ε4 allele, a genetic variant associated with higher ADRD risk.⁹

The study by Jimenez et al. has several advantages—prospective cohort design, large sample size, long and repeated follow-ups, and strong statistical analyses—that contribute to the robustness of their effect estimates. The findings from this study are generally consistent

with the existing evidence in this research field.^{10–13} There have been several epidemiological studies on greenspace and cognition in midlife and older adults.^{14,15} Most of these prior studies were cross-sectional in design and were conducted in Europe and North America. Their greenspace assessments were heterogeneous, with the studies adopting different metrics (e.g., NDVI, percentage of greenspace, walking in nature, gardening activity). Cognitive function also was measured by various methods, which limits direct comparisons between studies. Nevertheless, in line with the results reported by Jimenez et al., the prior studies supported beneficial associations between greater greenspace exposures and better cognitive function. Hence, enhancing green infrastructure should be implemented as an effective strategy to prevent cognition decline, especially in regions where greenspace levels are relatively lower.^{16,17}

In addition, the study by Jimenez et al. identified people who may derive more benefit from greenspace exposures. Specifically, people living in areas with lower socioeconomic status and higher population density seemed to derive a greater cognitive health benefit from greenspace. This indicates that, as the theory of equigenic environments states,¹⁸ enhancement of greenspace in poorer communities is a potential strategy for reducing socioeconomic health inequities, especially for cognitive health. Because areas with high population density generally have a large amount of built environment but less greenspace, this highlights the great importance and necessity of planning greenspace in urban regions. However, it is also possible that the existing built environment can be retrofitted with new better-designed greenspace, including areas on balconies and tops of buildings, which may better promote cognitive health. The authors also detected a stronger protective association between greenspace and cognitive decline among those with the APOE-ε4 allele, compared with those without the allele. This is a novel finding, and its underlying biological mechanisms are still not clear. Future studies should therefore seek to validate the finding and elucidate the exact mechanisms of the interaction.

The study points to several different approaches that investigators might consider in future studies. First, although NDVI is a widely used metric in greenspace–health investigations, it falls short of capturing the qualitative aspects and information on types, construction, and species of vegetation, which limits our ability to distinguish the health effects of different characteristics of greenspace. In addition, actual usage of greenspace often is not controlled for, which may compromise the precision in greenspace exposure assessment. Second, as the authors mention, greenspace may benefit cognition via restoration of attention, reduction of stress and air pollution, or as a means to increase physical activity, all of which could be explored in terms of their potential mediating roles.

Therefore, further studies should comprehensively assess greenspace exposures by integrating various metrics that represent different functions of these spaces and conduct dynamic monitoring of personal activity locations to assess exposure status based on personal activity trajectories throughout the research period. In addition, precise mechanisms underlying greenspace and cognitive function, such as reducing air pollution and noise,¹⁹

Address correspondence to Bo-Yi Yang, Joint International Research Laboratory of Environment and Health, Ministry of Education, Department of Occupational and Environmental Health, School of Public Health, Sun Yat-sen University, Guangzhou 510080, China. Email: yangby23@mail.sysu.edu.cn

The authors declare they have no conflicts of interest related to this work to disclose.

Conclusions and opinions are those of the individual authors and do not necessarily reflect the policies or views of EHP Publishing or the National Institute of Environmental Health Sciences.

Received 2 March 2024; Revised 19 April 2024; Accepted 14 June 2024; Published 17 July 2024.

Note to readers with disabilities: EHP strives to ensure that all journal content is accessible to all readers. However, some figures and Supplemental Material published in EHP articles may not conform to 508 standards due to the complexity of the information being presented. If you need assistance accessing journal content, please contact ehpsubmissions@niehs.nih.gov. Our staff will work with you to assess and meet your accessibility needs within 3 working days.

restoring attention,²⁰ mitigating stress,²¹ enhancing physical activity²² and social cohesion,²³ and increasing the diversity of microorganisms²⁴ should be explored by performing experimental studies or, at least, incorporating mediation analysis in epidemiological studies. Finally, considering that prior studies are mainly cross-sectional and cohort studies, we believe intervention studies should be carried out to provide the most robust epidemiological evidence possible.

References

- Innes KE, Sambamoorthi U. 2020. The association of osteoarthritis and related pain burden to incident Alzheimer's disease and related dementias: a retrospective cohort study of U.S. Medicare beneficiaries. *J Alzheimers Dis* 75(3):789–805, PMID: 32333589, <https://doi.org/10.3233/JAD-191311>.
- Flicker C, Ferris SH, Reisberg B. 1991. Mild cognitive impairment in the elderly: predictors of dementia. *Neurology* 41(7):1006–1009, PMID: 2067629, <https://doi.org/10.1212/wnl.41.7.1006>.
- Wilson RS, Segawa E, Boyle PA, Anagnos SE, Hizez LP, Bennett DA, et al. 2012. The natural history of cognitive decline in Alzheimer's disease. *Psychol Aging* 27(4):1008–1017, PMID: 22946521, <https://doi.org/10.1037/a0029857>.
- Yang BY, Zhao T, Hu LX, Browning MHEM, Heinrich J, Dharmage SC, et al. 2021. Greenspace and human health: an umbrella review. *Innovation (Camb)* 2(4):100164, PMID: 34622241, <https://doi.org/10.1016/j.xinn.2021.100164>.
- Lai KY, Kumari S, Gallacher J, Webster C, Sarkar C. 2024. Nexus between residential air pollution and physiological stress is moderated by greenness. *Nat Cities* 1(3):225–237, <https://doi.org/10.1038/s44284-024-00036-6>.
- Dadvand P, Nieuwenhuijsen MJ, Esnaola M, Fornes J, Basagaña X, Alvarez-Pedrerol M, et al. 2015. Green spaces and cognitive development in primary schoolchildren. *Proc Natl Acad Sci USA* 112(26):7937–7942, PMID: 26080420, <https://doi.org/10.1073/pnas.1503402112>.
- de Keijzer C, Gascon M, Nieuwenhuijsen MJ, Dadvand P. 2016. Long-term green space exposure and cognition across the life course: a systematic review. *Curr Environ Health Rep* 3(4):468–477, PMID: 27730509, <https://doi.org/10.1007/s40572-016-0116-x>.
- Jimenez MP, Wagner M, Laden F, Hart JE, Grodstein F, James P. 2024. Midlife residential greenness and late-life cognitive decline among Nurses' Health Study participants. *Environ Health Perspect* 132(7):077003, <https://doi.org/10.1289/EHP13588>.
- Zhu A, Yan L, Shu C, Zeng Y, Ji JS. 2020. APOE ε4 modifies effect of residential greenness on cognitive function among older adults: a longitudinal analysis in China. *Sci Rep* 10(1):82, PMID: 31919381, <https://doi.org/10.1038/s41598-019-57082-7>.
- Jimenez MP, Elliott EG, DeVille NV, Laden F, Hart JE, Weuve J, et al. 2022. Residential green space and cognitive function in a large cohort of middle-aged women. *JAMA Netw Open* 5(4):e229306, PMID: 35476063, <https://doi.org/10.1001/jamanetworkopen.2022.9306>.
- de Keijzer C, Tonne C, Basagaña X, Valentín A, Singh-Manoux A, Alonso J, et al. 2018. Residential surrounding greenness and cognitive decline: a 10-year follow-up of the Whitehall II cohort. *Environ Health Perspect* 126(7):077003, PMID: 30028296, <https://doi.org/10.1289/EHP2875>.
- Besser L, Galvin JE, Rodriguez D, Seeman T, Kukull W, Rapp SR, et al. 2019. Associations between neighborhood built environment and cognition vary by apolipoprotein E genotype: Multi-Ethnic Study of Atherosclerosis. *Health Place* 60:102188, PMID: 31797769, <https://doi.org/10.1016/j.healthplace.2019.102188>.
- Besser LM, Chang LC, Evenson KR, Hirsch JA, Michael YL, Galvin JE, et al. 2021. Associations between neighborhood park access and longitudinal change in cognition in older adults: the Multi-Ethnic Study of Atherosclerosis. *J Alzheimers Dis* 82(1):221–233, PMID: 34024841, <https://doi.org/10.3233/JAD-210370>.
- Zhao YL, Qu Y, Ou YN, Zhang YR, Tan L, Yu JT, et al. 2021. Environmental factors and risks of cognitive impairment and dementia: a systematic review and meta-analysis. *Ageing Res Rev* 72:101504, PMID: 34755643, <https://doi.org/10.1016/j.arr.2021.101504>.
- Chen X, Lee C, Huang H. 2022. Neighborhood built environment associated with cognition and dementia risk among older adults: a systematic literature review. *Soc Sci Med* 292:114560, PMID: 34776284, <https://doi.org/10.1016/j.socscimed.2021.114560>.
- Kleinschroth F, Savilaakso S, Kowarik I, Martínez PJ, Chang Y, Jakstis K, et al. 2024. Global disparities in urban green space use during the COVID-19 pandemic from a systematic review. *Nat Cities* 1(2):136–149, <https://doi.org/10.1038/s44284-023-00020-6>.
- Chen B, Wu S, Song Y, Webster C, Xu B, Gong P, et al. 2022. Contrasting inequality in human exposure to greenspace between cities of Global North and Global South. *Nat Commun* 13(1):4636, PMID: 35941122, <https://doi.org/10.1038/s41467-022-32258-4>.
- Mitchell R, Popham F. 2008. Effect of exposure to natural environment on health inequalities: an observational population study. *Lancet* 372(9650):1655–1660, PMID: 18994663, [https://doi.org/10.1016/S0140-6736\(08\)61689-X](https://doi.org/10.1016/S0140-6736(08)61689-X).
- Blanco MN, Shaffer RM, Li G, Adar SD, Carone M, Szpiro AA, et al. 2024. Traffic-related air pollution and dementia incidence in the Adult Changes in Thought Study. *Environ Int* 183:108418, PMID: 38185046, <https://doi.org/10.1016/j.envint.2024.108418>.
- Smart CM, Segalowitz SJ, Mulligan BP, MacDonald SWS. 2014. Attention capacity and self-report of subjective cognitive decline: a P3 ERP study. *Biol Psychol* 103:144–151, PMID: 25204705, <https://doi.org/10.1016/j.biopsycho.2014.08.016>.
- Ouanes S, Castela E, Gebreab S, von Gunten A, Preisig M, Popp J, et al. 2017. Life events, salivary cortisol, and cognitive performance in nondemented subjects: a population-based study. *Neurobiol Aging* 51:1–8, PMID: 28012996, <https://doi.org/10.1016/j.neurobiolaging.2016.11.014>.
- Najar J, Östling S, Gudmundsson P, Sundh V, Johansson L, Kern S, et al. 2019. Cognitive and physical activity and dementia: a 44-year longitudinal population study of women. *Neurology* 92(12):e1322–e1330, PMID: 30787164, <https://doi.org/10.1212/WNL.0000000000007021>.
- Zhang W, Liu S, Sun F, Dong X. 2019. Neighborhood social cohesion and cognitive function in U.S. Chinese older adults—findings from the PINE study. *Ageing Ment Health* 23(9):1113–1121, PMID: 30518241, <https://doi.org/10.1080/13607863.2018.1480705>.
- Zhang YD, Fan SJ, Zhang Z, Li JX, Liu XX, Hu LX, et al. 2023. Association between residential greenness and human microbiota: evidence from multiple countries. *Environ Health Perspect* 131(8):087010, PMID: 37585351, <https://doi.org/10.1289/EHP12186>.